Uniformly accelerated rectilinear motion

Video lesson companion (300 s)

Content version: v2 • translation pending teacher review

Goal

After the video, the learner will be able to compute the velocity and displacement of a uniformly accelerated body from its initial velocity, acceleration and time, and to check the answer by the area under the v(t) graph.

Brief explanation

- A car travels at 2 m/s and accelerates; after 4 s its speed is 14 m/s. How many metres did it cover? We will find the answer two ways.
- If speed never changes, motion is uniform ($v = s/t$). Here speed grows by the same amount each second: +3 m/s every second — so acceleration $a = 3$ m/s^2 and the motion is uniformly accelerated.
- Acceleration: $a = \Delta v/\Delta t$; unit — m/s^2. The Ox axis points along the motion; everything forward takes the plus sign.
- Velocity: $v = v_0 + at = 2 + 3\cdot4 = 14$ m/s. Unit check: (m/s^2)·s = m/s.
- Displacement: $s = v_0t + at^2/2 = 2\cdot4 + 3\cdot16/2 = 8 + 24 = 32$ m. The first term is the distance without acceleration, the second is the gain from it.
- The v(t) graph is the line $v = 2 + 3t$ through (0; 2) and (4; 14); the slope of the line equals the acceleration a.
- The area under the v(t) graph numerically equals the displacement: trapezoid $(2 + 14)/2\cdot4 = 32$ m — matches the formula.
- Traps: $14\cdot4 = 56$ is wrong (speed was growing; the average is needed); slope is acceleration, area is displacement.

Applicability

- rectilinear: motion along a straight line;
- constant acceleration: acceleration magnitude and direction do not change;
- velocity sign: velocity keeps its sign, so displacement numerically equals distance travelled;
- SI units: all quantities in SI (m, s, m/s, m/s^2).

Ox directed along the motion; forward projections take the + sign (TYҚ case). TKҚ and direction reversal are out of lesson scope.

Quantities and units

Symbol	Quantity	Value	Status
v_0	initial velocity	2 m/s	input
a	acceleration	3 m/s^2	input
t	time	4 s	input
v	final velocity	14 m/s	derived (Stage G)
s	displacement	32 m	derived (Stage G)
v_0	initial velocity (pause task B6)	0 m/s	input
a	acceleration (B6)	2 m/s^2	input
t	time (B6)	5 s	input
s	displacement (B6)	25 m	derived (Stage G)

Formulas and conditions

definition of acceleration:

$$a = \Delta v / \Delta t$$

definition; always true.

velocity:

$$v = v_0 + at$$

only for constant acceleration (conditions above).

displacement:

$$s = v_0 t + a t^2 / 2$$

only for constant acceleration (conditions above).

displacement via average velocity:

$$s = (v_0 + v) t / 2$$

only for constant acceleration (conditions above).

Figures

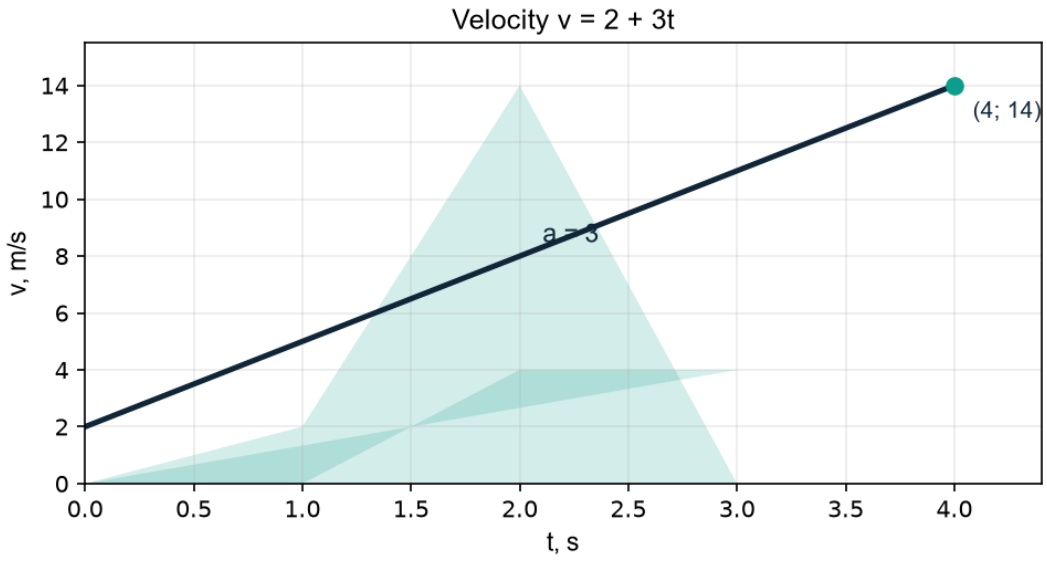


Figure 1: Velocity graph v = 2 + 3t. The slope of the line equals the acceleration (a = 3 m/s^2); the shaded area under the graph numerically equals the displacement (s = 32 m).

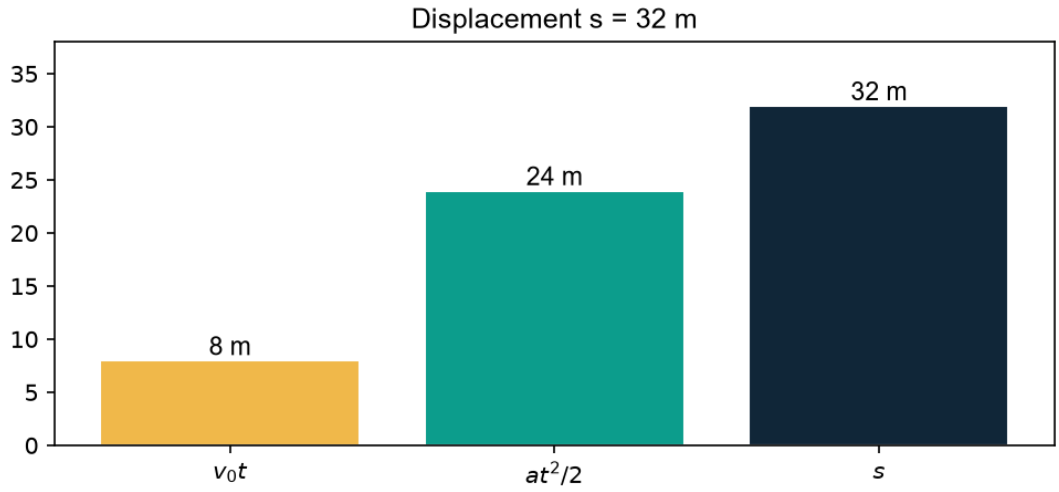


Figure 2: Displacement terms: v_0t = 8 m (no acceleration) and $at^2/2$ = 24 m (gain from acceleration); total s = 32 m.

Worked example

Given: v_0 = 2 m/s, a = 3 m/s^2, t = 4 s. Find v and s.

1. Velocity: $v = v_0 + at = 2 + 3{\cdot}4 = 14$ m/s.
2. Displacement: $s = v_0t + at^2/2 = 2{\cdot}4 + 3{\cdot}16/2 = 8 + 24 = 32$ m.
3. Area check: $s = (v_0 + v)t/2 = (2 + 14){\cdot}4/2 = 32$ m. It matches.

Pause task (B6): $v_0 = 0$, a = 2 m/s^2, t = 5 s $\rightarrow$ $s = 0{\cdot}5 + 2{\cdot}25/2 = 25$ m.

Self-study tasks

Solve on your own; answers are in a separate section.

H1. An object starts at 2 m/s and accelerates at 3 m/s^2. What is its speed after 4 seconds?

A. 14 m/s
B. 12 m/s
C. 9 m/s
D. 6 m/s

H2. An object moves with initial speed 2 m/s and acceleration 3 m/s^2 without changing direction. Find its displacement in 4 s.

A. 32 m
B. 56 m
C. 24 m
D. 20 m

H3. Which motion is called uniformly accelerated?

A. Velocity changes by equal amounts in equal times
B. Velocity stays constant
C. Acceleration changes over time
D. Distance is the same in equal times

H4. What does the slope of the line on a velocity–time graph show?

A. Acceleration
B. Displacement
C. Velocity
D. Time

H5. On a v(t) graph a straight line runs from (0 s; 2 m/s) to (4 s; 14 m/s). Find the displacement.

A. 32 m

B. 48 m

C. 56 m

D. 24 m

Answers

Answers with short explanations. **H1 — A** (14 m/s). Acceleration adds 3 m/s of speed each second. So $v = v_0 + at = 2 + 3 \cdot 4 = 14$ m/s. **H2 — A** (32 m). $s = v_0 t + at^2/2 = 2 \cdot 4 + 3 \cdot 16/2 = 8 + 24 = 32$ m. 56 m comes from plugging the final speed into the uniform-motion formula. **H3 — A** (Velocity changes by equal amounts in equal times). In uniformly accelerated motion velocity gains equal increments in equal times while acceleration stays constant (Zakirova 9, §3). D describes uniform motion. **H4 — A** (Acceleration). The slope of v(t) equals acceleration: $a = \Delta v/\Delta t$. Displacement is the area under the graph, not the slope (Zakirova 10, §3). **H5 — A** (32 m). Trapezoid area: $(2 + 14)/2 \cdot 4 = 32$ m. 24 m is the triangle alone, without the rectangle.

Sources

- Zakirova, Physics, grade 9, §3 — uniformly accelerated motion, v(t) graphs (PDF pp. 19–26).
- Zakirova, Physics, grade 10, §3 — kinematics equations, v(t) slope and area (PDF pp. 15–18, 20).
- Zakirova, Physics, grade 11 — notation cross-check (PDF pp. 8–12).
- “369 formulas” — formula cross-check (PDF pp. 6–7).
- FBS — notation and unit cross-check (PDF pp. 6–7).
- ENT collections: `https://testcenter.kz`

Sources were used for cross-checking only; no textbook text, figure, or exercise is reproduced.